

2D Navier–Stokes recurrent adversarial attack

(20 frames: $x_0, \dots, x_{19}$; input window 0–9, recurrent prediction 10–19)

—→ $\mathcal{G}$ forward

—→ $\mathcal{F}_\theta$ forward

—→ $\mathcal{F}_\theta$ backward

- - - - -→ $\mathcal{G}$ backward (detach: no backprop)

$$x_0^\delta = x_0 + \delta_k, \quad x_j^\mathcal{G} = \mathcal{G}_j(x_0^\delta), \quad j = 1, \dots, 19$$

$$[\hat{x}_{10}^\mathcal{F}, \dots, \hat{x}_{19}^\mathcal{F}] = \mathcal{F}_\theta^{\text{rec}}([x_0^\delta, x_1^\mathcal{G}, \dots, x_9^\mathcal{G}]), \quad \mathcal{L}_{\text{adv}}(\delta_k) = \|\hat{x}_{19}^\mathcal{F} - x_{19}^\mathcal{G}\|_2^2$$

mode_spec = $(c_1, \dots, c_9, c_{19})$, $c_j \in \{A, D, W\}$

A: dictionary/approx.; D: detached solver; W: solver backprop

